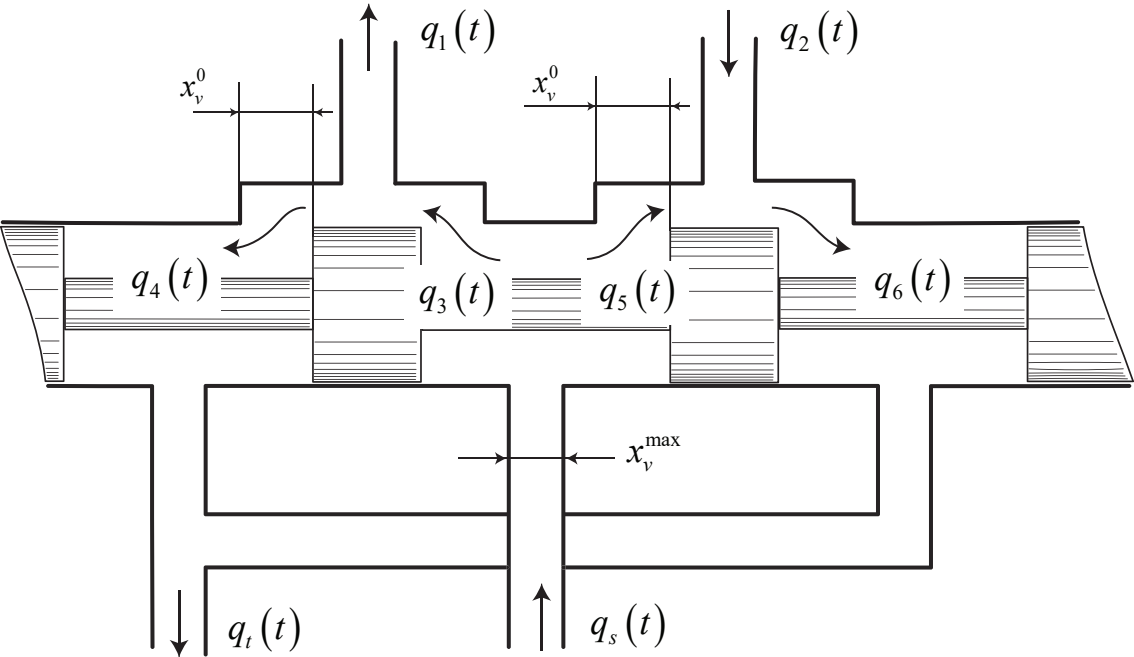

$q_1(t)$
$q_2(t)$
x_v^0
x_v^0
$q_4(t)$
$q_3(t)$
$q_5(t)$
$q_6(t)$
$x_v^{\max}$
$q_t(t)$
$q_s(t)$